Data Codebook

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Data Dictionary: Summary of Variables

Note: Data structure was automatically generated from DataFrame analysis. Variable metadata may need review and updating for accuracy.

Variable Name	Data Type	Length	Categorical	Variable Label
id	Int			Variable Label Missing
name	String			Variable Label Missing
score	Int			Variable Label Missing

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Variable Details and Notes

The following pages provide details on each variable. Where applicable, notes provide links to verify data. Categorical variables include details on category codes.

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id: Variable Label Missing

Variable characteristic	Variable details
variable type	numeric (Int)
total cases	3
valid cases	3
missing cases	0
unit of measure	Unknown
unit of analysis	Unknown
range	minimum value: 1.00 to maximum value: 3.00
mean	2.00
median	2.00
standard deviation	1.00
10th percentile	1.20
25th percentile	1.50
50th percentile	2.00
75th percentile	2.50
90th percentile	2.80

Variable Notes: id

WARNING: Auto-generated data structure - please review and update metadata as needed.

Data Codebook

name: Variable Label Missing

Variable characteristic	Variable details
variable type	string
total cases	3
valid cases	3
missing cases	0
unit of measure	Unknown
unit of analysis	Unknown
unique values	3
minimum length	5
maximum length	7
example 1	Alice
example 2	Bob
example 3	Charlie
example 4	Alice

Variable Notes: name

WARNING: Auto-generated data structure - please review and update metadata as needed.

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score: Variable Label Missing

Variable characteristic	Variable details
variable type	numeric (Int)
total cases	3
valid cases	3
missing cases	0
unit of measure	Unknown
unit of analysis	Unknown
range	minimum value: 78.00 to maximum value: 92.00
mean	85.00
median	85.00
standard deviation	7.00
10th percentile	79.40
25th percentile	81.50
50th percentile	85.00
75th percentile	88.50
90th percentile	90.60

Variable Notes: score

WARNING: Auto-generated data structure - please review and update metadata as needed.

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Instructions for Adding Project Overview

Instructions for Project Overview

A project overview provides essential context that helps users understand the purpose, scope, and key details of your dataset. It orients new users, supports transparency, and ensures that the data is used appropriately and interpreted correctly.

Please include information about your project, such as:

Required:
Project title: Clear, descriptive name for your project or dataset
Principal Investigator(s): Lead researcher(s) with contact information

Good to have:
Required citations: How others should cite your work or dataset
Project description: What questions is this research trying to answer and why it matters
Funding source: Grant numbers, award names, and funding agency information
Data provenance: Detailed information about data sources, collection methods, and any synthetic/simulated elements

Good to include if survey data:
Data collection context - When, where, and how the data was collected
Target population - Who or what the data represents
Sample size and methodology - How many observations and what sampling approach
Time period - When the data was collected and what time frame it covers
Geographic scope - Location(s) where the data applies

Good to include if known:
Related works - Publications, reports, or other research using or related to this data
Keywords - Terms that help others discover and categorize your work
License - Legal terms governing how others may use your data
Limitations and considerations - Important caveats about data use and interpretation

File Format Requirements

When you have your project overview file completed, pypdfcodebook requires that this is saved as a **markdown file** (with `.md` extension). Markdown is a text file with simple formatting options.

Formatting Support in PDF Output

Important: pypdfcodebook uses fpdf2 to convert your markdown content to PDF, which has limited formatting support. Only these markdown features will be rendered in the final PDF:

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- Code blocks or inline code - these become plain text

Recommendations for Best Results

1. **Use descriptive text** rather than relying on headers for organization
2. **Use bold formatting** (`**text**`) to emphasize important information like section names
3. **Write in paragraph form** rather than using bullet lists, since lists won't format properly
4. **Keep it simple** - focus on clear, well-written content rather than complex formatting

Creating Your Files

For both project overview and key terms files:
- Save as `.md` files (e.g., `project_overview.md`, `keyterms.md`)
- Use any text editor (Notepad, VS Code, etc.)
- Focus on clear content over complex formatting
- Use the simple bold/italic formatting sparingly for emphasis

This section was auto-generated using the default instruction mode in pypdfcodebook. To turn off instruction mode set instruction_mode to False in the codebook command. Please replace with actual project details relevant to your dataset and research.

Data Codebook

Instructions for Adding Data Structure File

Instructions for Data Structure File

A data structure file defines the metadata and specifications for each variable in your dataset. This Python dictionary provides essential information that pypdfcodebook uses to create comprehensive variable documentation in your PDF codebook.

Please create a DATA_STRUCTURE dictionary following this format:

```
DATA_STRUCTURE = {
    'variable_name' : {
        Required fields for all variables:
        'label' : 'Human-readable variable description',
        'DataType' : 'String|Int|Float|Bool',  Data type as it appears in your data
        'pyType' : str|int|float|bool|"category",  Python type for processing
        'AnalysisUnit' : 'What each row represents (e.g., Survey response, Person, etc.)',
        'MeasureUnit' : 'Units of measurement (e.g., Years, Dollars, Responses, etc.)',

        Optional but recommended:
        'notes' : 'Additional context, methodology, or important details about this variable',

        Required for categorical variables:
        'categorical' : True,  Set to True if this is a categorical variable
        'categorical_type' : 'nominal|ordinal',  Type of categorical variable
        'categories_dict' : {
            1 : '1. First category label',
            2 : '2. Second category label',
            ... continue for all categories
        },
        'categories' : [
            '1. First category label',
            '2. Second category label',
            ... list matching the categories_dict
        ],

        Optional for weighted analysis:
        'primary_key' : 'variable_name_of_unique_identifier',
        'weight_var' : 'variable_name_of_weight_variable',
    },

    Add additional variables following the same structure...
}
```

Example based on sample survey data:

```
DATA_STRUCTURE = {
    'rid': {
        'label': 'Random Survey Response ID',
        'DataType': 'String',
        'pyType': str,
        'AnalysisUnit': 'Survey response',
        'MeasureUnit': 'Responses',
        'notes': 'join([
            '1. Unique, non-missing key for sample data.',
```

```
            '2. Randomly generated 5 digit alphanumeric string identifier.',
            '3. Range from A1000 to Z9999.'
        ])
    },
    'region': {
        'label': 'Geographic Region',
        'DataType': 'Int',
        'pyType': "category",
        'categorical': True,
        'categorical_type': 'nominal',
        'AnalysisUnit': 'Survey response',
        'MeasureUnit': 'Responses by Region',
        'categories_dict': {
            1: '1. North',
            2: '2. South',
            3: '3. East',
            4: '4. West'
        },
        'categories': [
            '1. North',
            '2. South',
            '3. East',
            '4. West'
        ],
        'primary_key': 'rid',
        'weight_var': 'weight'
    },
    'age': {
        'label': 'Age in Years',
        'DataType': 'Int',
        'pyType': int,
        'AnalysisUnit': 'Survey response',
        'MeasureUnit': 'Years',
        'notes': 'join([
            '1. Age of survey respondent at time of survey.',
            '2. Range from 18 to 99 years.',
            '3. Missing values indicate participant did not provide age information.',
            '4. Surveys completed between November 2025 and December 2025.'
        ]),
        'primary_key': 'rid',
        'weight_var': 'weight'
    }
}
```

Field Specifications

Required Fields (all variables): label is a clear, descriptive name for the variable. **DataType** is the data type as stored ('String', 'Int', 'Float', 'Bool'). **pyType** is the Python type for processing (str, int, float, bool, or "category"). **AnalysisUnit** describes what each observation represents. **MeasureUnit** specifies units or scale of measurement.

Categorical Variables: Set **categorical** to True. Use **categorical_type** as 'nominal' for unordered categories, 'ordinal' for ordered categories. **categories_dict** should be a dictionary mapping numeric codes to labeled categories. **categories** should be a list of category labels matching the dictionary.

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Optional but Recommended: notes provides additional context, methodology notes, or important details. **primary_key** names the unique identifier variable (for analysis). **weight_var** names the weight variable (for weighted analysis).

Data Types Guide

'String' is for text data (names, IDs, open responses). **'Int'** is for whole numbers (ages, counts, categorical codes). **'Float'** is for decimal numbers (weights, measurements, percentages). **'Bool'** is for True/False values. **pyType "category"** should be used for categorical variables regardless of underlying storage type.

File Requirements

Save as Python file using .py extension (e.g., data_structure.py). **Variable naming** must use the exact column names from your dataset. **Dictionary format** must be valid Python dictionary syntax. **Category consistency** requires that categories_dict and categories list match exactly.

Notes Format

For multi-line notes, use the join pattern: 'notes': 'join(['1. First important point.', '2. Second important point.', '3. Additional context or methodology.'])

Creating Your Data Structure File

Create a new Python file with .py extension. Define your DATA_STRUCTURE dictionary using the exact variable names from your dataset. For each variable, include all required fields and any relevant optional fields. For categorical variables, make sure to include the categorical-specific fields. Save the file in your project directory where pypdfcodebook can access it.

Common Patterns

For survey data, typically include a unique identifier (like rid), demographic variables (like age, region), response variables (like satisfaction scores), and any weight variables for analysis. For experimental data, include treatment indicators, outcome measures, and control variables. For observational data, include all measured variables with appropriate data types and units.

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Data Codebook

<u>Instructions for Adding Key Terms and Definitions</u>

Instructions for Key Terms and Definitions

Key terms provide essential definitions that ensure future data users do not have to assume what terms mean, promoting clear understanding and consistent interpretation of the dataset and its documentation.

Please define key terms related to your project, including:

- **Technical terminology** used in variable names or descriptions
- **Domain-specific concepts** relevant to your field of study
- **Measurement units** and their definitions
- **Categorical codes** and what they represent
- **Analysis concepts** that may not be familiar to all users

This process may require a literature review and communication with the data creator. If you do not know what a term in the dataset means it is ok to simply state "Definition unknown".

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